Atlas2350 Rev A | 80 x 75mm, four layers, 1.6mm FR-4, ENIG
All components top-side; PCBWay includes SMT and through-hole connectors.
Fabrication: 0.127mm clearance, 0.15mm signal width, 0.30mm drills / 0.60mm vias.
Filled/capped through vias required at QFN exposed pads and module thermal pad.

MCU

RP2350A and external 1.1V core supply

mcu.kicad_sch

Memory

4MiB QSPI NOR and 8MiB QSPI PSRAM

memory.kicad_sch

Power

USB-controlled LiPo charging and 3.3V power path

power.kicad_sch

Interfaces

USB-C, microSD, Qwiic and addressable RGB

interfaces.kicad_sch

Wireless

Wi-Fi / BLE UART coprocessor and programming

wireless.kicad_sch

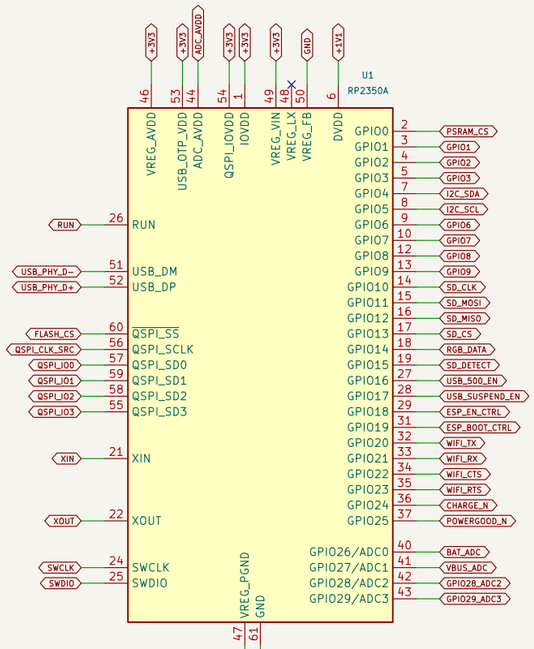
DECISIONS
3.3V buck-boost powers MCU, memories, SD and radio from USB or LiPo.
External 1.1V LDO avoids the RP2350 internal buck magnetic-polarity/layout dependency.
External 3-wire protected battery required: BAT+, GND, 10k NTC. No battery or microSD card in assembly BOM.
Wi-Fi/BLE is an ESP32-C3 coprocessor, not native RP2350 functionality; firmware required.
Factory assemble every BOM row. Mechanical holes and bare test pads are not components.
No DNI/DNP component options. See README and fab/assembly-notes.md before ordering.

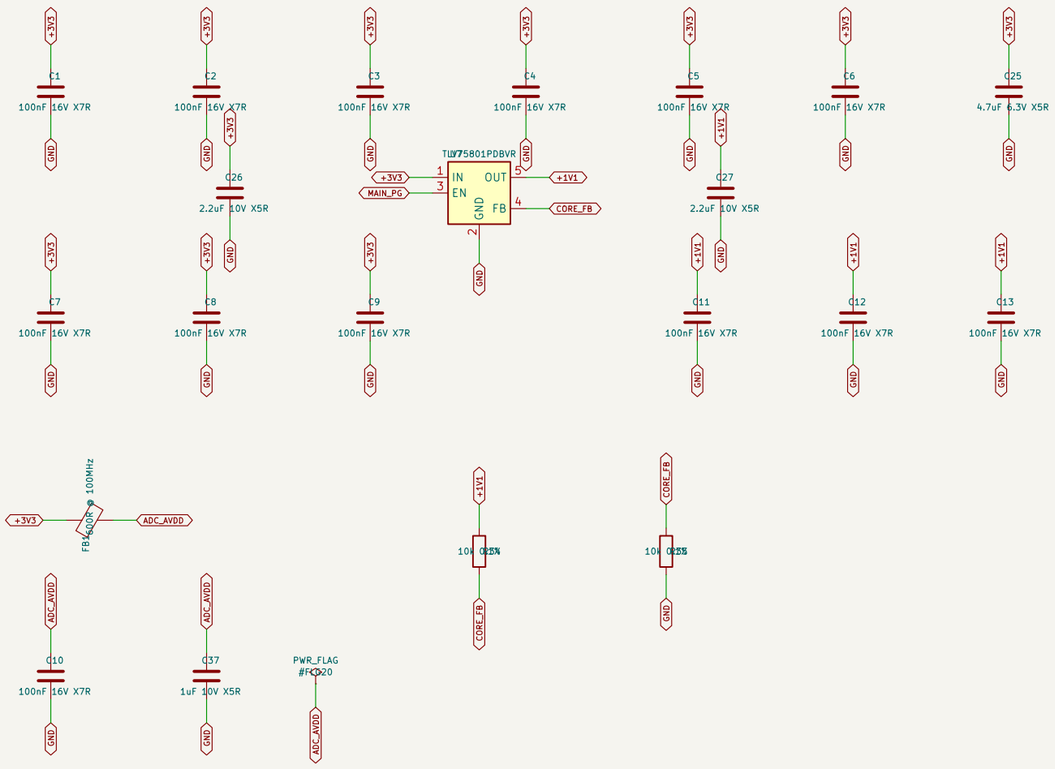
RP2350A + QSPI flash/PSRAM + Wi-Fi/BLE + USB-C + LiPo
Atlas2350
Sheet: /
File: Atlas2350.kicad_sch

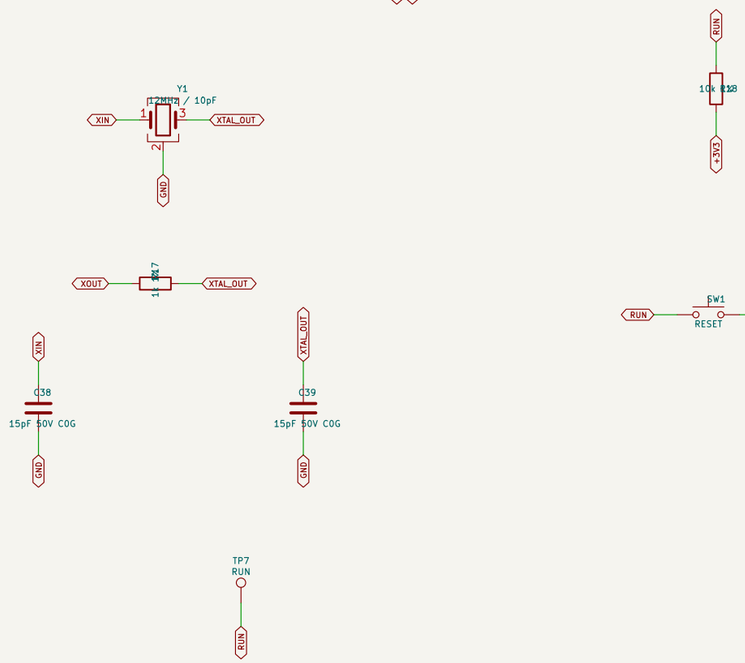
Title: Atlas2350 connected LiPo development board

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.99.0-6305-g5ac898839d		Id: 1/6

150MHz nominal. External LDO supplies DVDD: VREG_FB=GND, LX=NC.
Keep VREG_VIN and VREG_AVDD powered. Set VREG.HIZ=1 in firmware; never select low-power VREG mode.





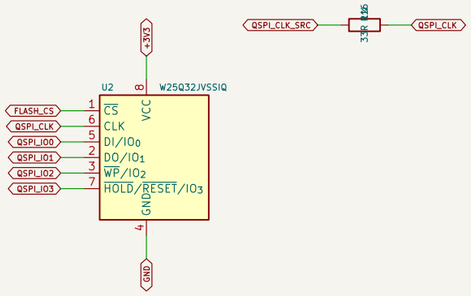


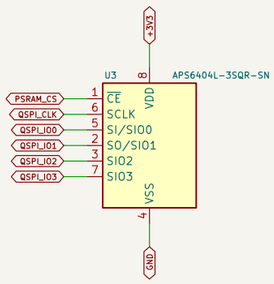
IOs are 3.3V, not 5V tolerant. No SWO.
Expansion pin19 VBUS and pin20 VSYS are outputs only: never back-power.

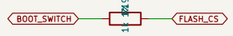
PCBWay full assembly; all fitted parts specified by MPN
Atlas2350

Sheet: /MCU/		
File: mcu.kicad_sch		
Title: RP2350A and external 1.1V core supply		
Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.99.0-6305-g5ac898839d		Id: 2/6

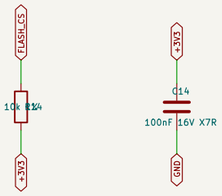
QSPI bus shared; CS0 is flash, GP0 is PSRAM CS1. Both chip selects pulled high.
33R source termination on clock. Start QSPI <=62.5MHz; validate signal integrity on first articles.

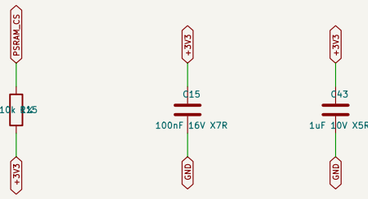












BOOTSEL: hold SW2, pulse RUN (SW1), release SW2 after USB enumerates.
1k limits current if BOOTSEL is pressed while the flash chip-select output is active.

PCBWay full assembly; all fitted parts specified by MPN

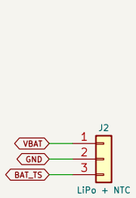
Atlas2350

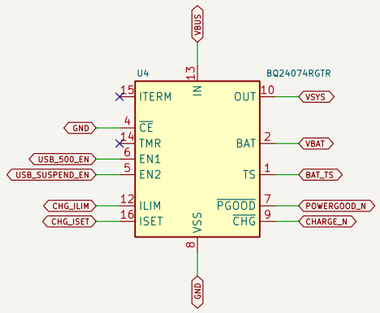
Sheet: /Memory/
File: memory.kicad_sch

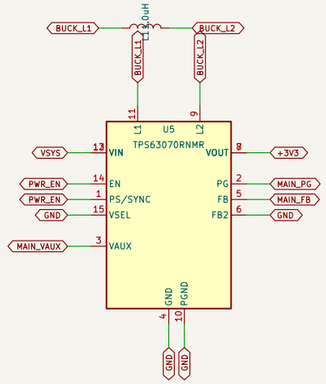
Title: 4MiB QSPI NOR and 8MiB QSPI PSRAM

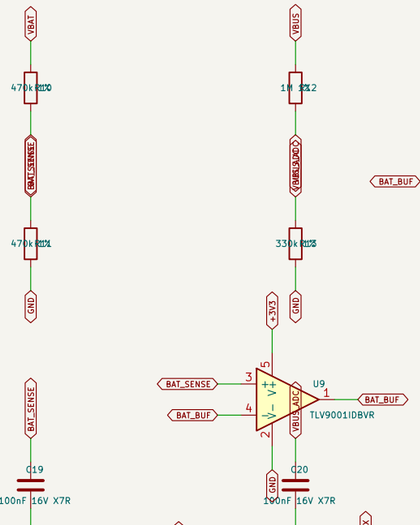
Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.99.0-6305-g5ac898839d		Id: 3/6

USB_500_EN=0 at reset: 100mA USB mode. Set to1 only after a 500mA USB grant.
Charger ISET=3.57k: 250mA nominal. ITERM=NC: 10% termination; TMR=NC: 5h safety timer.









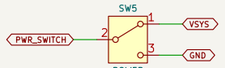


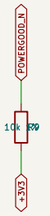




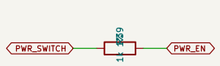


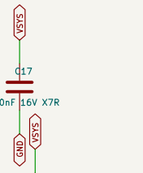












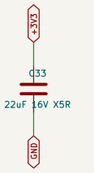












VBAT approximately 2 * BAT_ADC divider buffered by U9.
VBUS_ADC is presence detection, not precision voltage measurement.
USB_500_EN / USB_SUSPEND_EN: 00=100mA,10=500mA,11=suspend.

BATTERY: protected 1S LiPo, >=500mAh, 4.2V charging; 10k NTC required.
J2 pins 1=BAT+, 2=GND, 3=NTC. THIS IS A CUSTOM 3-PIN PINOUT.
No battery supplied or soldered. Missing NTC inhibits charging.

PCBWay full assembly; all fitted parts specified by MPN

Atlas2350

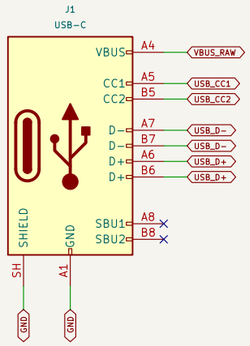
Sheet: /Power/
File: power.kicad_sch

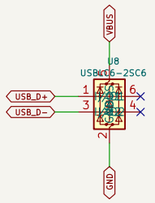
Title: USB-controlled LiPo charging and 3.3V power path

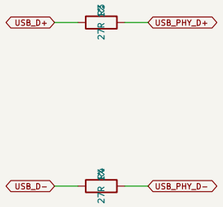
Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.99.0-6305-g5ac898839d		Id: 4/6

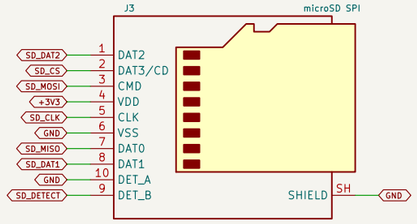
USB2.0 device / 5V sink only; no PD, no host power sourcing.
Both CC pins use separate 5.1k Rd. ESD array sits at the receptacle; 27R series resistors at U1.



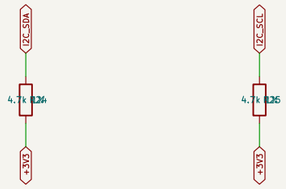


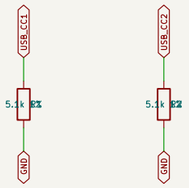


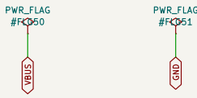


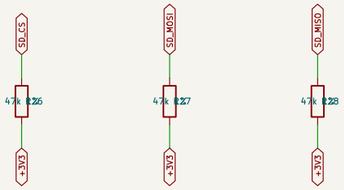


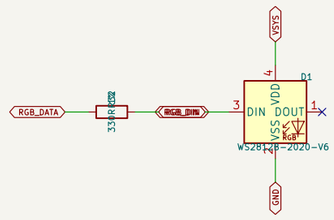


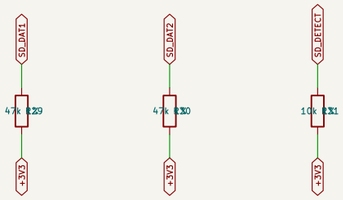


















RGB V6 only: VIH=0.55*VDD; powered from VSYS.
Blank RGB below VBAT about3.6V under load; limit current to15mA.
RGB not guaranteed when VSYS<3.3V. I2C 3.3V/4.7k; SD SPI initially<=25MHz.

PCBWay full assembly; all fitted parts specified by MPN

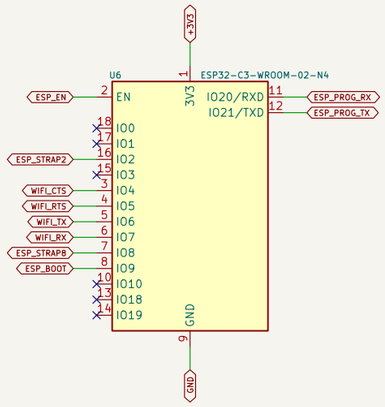
Atlas2350

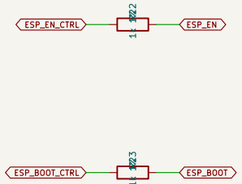
Sheet: /Interfaces/
File: interfaces.kicad_sch

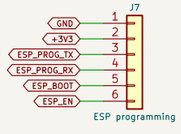
Title: USB-C, microSD, Qwiic and addressable RGB

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.99.0-6305-g5ac898839d		Id: 5/6

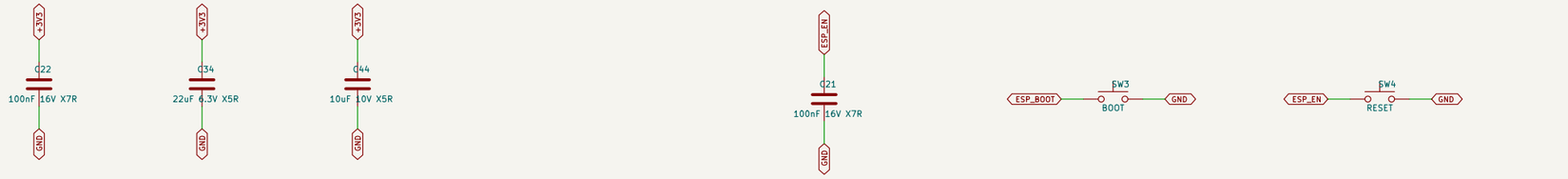
ESP-AT UART1: GPIO6 RX <- RP GP20; GPIO7 TX -> RP GP21.
GPIO4 RTS -> RP GP22 CTS; GPIO5 CTS <- RP GP23 RTS.
UART0 GPIO20/21 is a separate 3.3V programming interface; start115200baud.



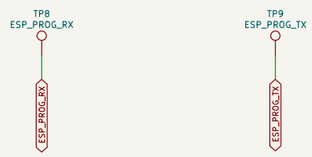








Radio disabled by 10k EN pull-down at cold boot. Enable only after power budget is known.
GP19 controls IO9 as open-drain: release for normal boot, pull low while pulsing EN for download.
IO8 and IO2 strap high. Antenna projects beyond top PCB edge; do not enclose with metal.



PCBWay full assembly; all fitted parts specified by MPN
Atlas2350

Sheet: /Wireless/
File: wireless.kicad_sch

Title: Wi-Fi / BLE UART coprocessor and programming

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.99.0-6305-g5ac898839d		Id: 6/6